devinhorowitz/solar-business-card

take a very close look at this project and determine whether we could also integrate other uses for the NFC chip built into it.

NFC Chip Integration Assessment

Based on the provided project data, I can see this is a repository for a "solar-business-card" containing a Bill of Materials (BOM), GitHub workflows, and documentation. However, the text provided is a raw binary dump of the project archive, which means the readable code, schematics, or specific NFC chip part numbers are not visible in this context.

ZIP +1

Despite not having the exact chip model, the straightforward answer is **yes, you can almost certainly integrate other uses into the NFC chip**, provided the hardware has available memory and is read/write capable.

Potential NFC Applications

Most smart business cards utilize standard NDEF (NFC Data Exchange Format) tags. If your embedded chip supports this, you can configure it for various alternative functions:

- **Dynamic Redirection:** Instead of a static vCard, program the chip with a URL pointing to a backend server you control. This allows you to change what the card "does" (e.g., pointing to a portfolio today, and a new product launch tomorrow) without reprogramming the physical card.
- **Digital Hubs:** Redirecting users to a Linktree, GitHub repository, or social media hub.
- **Wi-Fi Provisioning:** Automatically connecting the scanning device to a specific guest Wi-Fi network.
- **App Launching & Automations:** Triggering deep links that open specific apps, or initiating Apple Shortcuts and Android Tasker routines on the scanner's phone.

Hardware Considerations

To determine exactly how much functionality you can pack into the card, you will need to verify two things:

- **Memory Capacity:** Common chips like the NTAG213 hold 144 bytes, while the NTAG216 holds 888 bytes. If you want to store multiple records (like a vCard *plus* a URL), you will need a higher-capacity chip.
- **Lock Status:** You must ensure the chip's memory is not permanently write-locked by your current programming or manufacturing process.

Do you have the specific part number for the NFC chip from the Bill of Materials so we can verify its exact memory limits and capabilities?

CSV

solar-glow-drh-v4_0-bom

here is the bom

Hide code

Python

```
import pandas as pd

try:
    df = pd.read_csv("solar-glow-drh-v4_0-bom.csv")
    # Show columns to understand structure
    print("Columns:", df.columns.tolist())

    # Search for NFC related components
    nfc_keywords = ['NFC', 'RFID', 'NTAG', 'IC', 'Chip', 'Antenna', 'ST25']

    # filter rows where any column contains the keywords (case insensitive)
    mask = df.apply(lambda row: row.astype(str).str.contains('|'.join(nfc_keywords), ca

    print("\nPotential NFC Components:")
    print(df[mask].to_string())

except Exception as e:
    print(f"Error: {e}")
```

Code output

```
Columns: ['Row', 'Description', 'Part', 'References', 'Value', 'Footprint', 'Quantity F

Potential NFC Components:
               Row
0                1  NFC tank trim, 0805 high-Q C0G +/-2% (Johanson S-series, ultra-low
1                2
3                4
4                5
5                6
6                7
7                8
9               10
10              11
15              16
16              17
17              18
18              19
19              20
20              21
21              22
22              23
23              24
26      Schematic:
30  KiCad Version:
31     Statistics:
```

Awesome, thanks for providing the BOM! I ran a quick analysis on **solar-glow-drh-v4_0-bom.csv** and found exactly what we needed.

Your NFC chip is the **NXP NT3H2211W0FHKH**, which belongs to the **NTAG I2C Plus 2K** family. This is a fantastic choice. It is not just a standard passive NFC tag—it is a highly capable dual-interface chip designed to act as a bridge between an NFC reader (like a smartphone) and your onboard microcontroller (the AVR64EA28, which is also listed in your BOM).

Because of this architecture, your options for integrating other uses are vast.

Advanced Capabilities of the NT3H2211

- **Dual Interface (NFC & I2C):** The chip communicates wirelessly with a phone and physically with your onboard AVR microcontroller via the I2C bus.
- **Pass-Through Mode:** It features a 64-byte SRAM buffer that allows real-time, two-way data exchange between a scanning phone and your MCU without wearing out the EEPROM memory.
- **Energy Harvesting:** It can harvest power directly from the scanning phone's NFC field (up to roughly 15 mW). This could potentially wake up your MCU or power an LED even if the card's main energy storage is depleted.
- **Ample Storage:** With 2 kbit of memory (1912 bytes of user memory), you have plenty of room for multiple NDEF records, allowing you to store a vCard, a URL, and custom application data simultaneously.

Potential New Uses for Your Card

- **Two-Way Interactive Apps:** Instead of just sending a URL to a phone, a custom web app (using WebNFC) or native smartphone app can send commands *back* to the card. For instance, the scanner's phone could configure settings on the card or trigger specific LED glow patterns.
- **Dynamic Data Serving:** Because your MCU can write to the NFC tag's memory, the card can update its own NFC message dynamically. It could track and display how many times it has been tapped or serve real-time sensor data (such as ambient light levels or battery status) directly to the phone.
- **Secure Authentication:** The chip supports 32-bit password protection. You could utilize the card as a physical hardware key to authenticate access to a digital portfolio, a secure web portal, or a physical smart lock.

You have a very versatile hardware foundation here. Are you more interested in exploring the interactive features (like real-time, two-way communication with a phone) or the dynamic data features (where the card updates its own content)?

looking at the repo and the bom - make a proposal that the build team would read and consider for potential features to add on

Here is a proposal tailored specifically to the hardware footprint of your project, designed for the build team to review. It leverages the synergies between the components already present in your BOM.

Feature Expansion Proposal: Solar Glow Business Card v4.0

Executive Summary

The v4.0 hardware represents a highly capable embedded system. The inclusion of the **AEM10300** solar harvester, **ADXL367** accelerometer, **AVR64EA28** MCU, **512kbit FeRAM**, and the dual-interface **NT3H2211** NFC chip provides a foundation that extends far beyond a static digital business card. This proposal outlines three firmware-driven features that require zero hardware changes, maximizing the return on your existing component investments.

Proposed Features

1. Gesture-Based Profile Switching

Components Utilized: ADXL367 (Accelerometer), AVR64EA28 (MCU), NT3H2211 (NFC)
Concept: Currently, digital business cards usually share a single link. By utilizing the ultra-low-power ADXL367's tap/double-tap wake feature, users can cycle through different digital profiles before handing the card to someone to scan.

- **Execution:** A single tap configures the NFC NDEF record to point to a standard vCard/contact file. A double tap prompts the MCU to rewrite the NDEF record via I2C to point to a specific product demo, portfolio, or GitHub repository.
- **User Feedback:** The onboard LEDs can flash a specific color or pattern (e.g., 1 blink for Contact, 2 blinks for Portfolio) to confirm the active mode.

2. Environmental & Tap Data Logging (The "Card Analytics" Dashboard)

Components Utilized: AEM10300 (PMIC), MB85RC512TY (FeRAM), NT3H2211 (NFC SRAM Pass-through)
Concept: The 512kbit FeRAM offers massive, high-endurance storage. The card can act as a localized data logger, tracking its own environmental conditions and interaction metrics.

- **Execution:** The MCU periodically wakes to log solar harvesting levels (from the AEM10300) and timestamps of accelerometer wakes or NFC field detections.
- **Data Retrieval:** When the owner scans the card using a custom companion app or WebNFC page, the phone can query the MCU via the NT3H2211's 64-byte SRAM pass-through. The MCU reads the FeRAM and streams the analytics dashboard back to the phone, turning the card into a smart environmental sensor.

3. Zero-Battery "Vampire" Wake

Components Utilized: NT3H2211 (NFC Energy Harvesting), TPS22917 (Load Switch), LEDs
Concept: If the card has been stored in a dark drawer for months and the main energy buffer is completely depleted, the card can still provide a "wow" factor using the NFC scanner's RF field.

- **Execution:** The NT3H2211 can harvest up to ~15mW of power from a scanning smartphone. This harvested energy can be routed through the TPS22917 load switch to briefly wake the MCU or directly pulse the "solar-glow" LEDs.
- **Value:** Guarantees that the card is never truly "dead" when handed to a client, always delivering the digital payload and a visual LED response entirely powered by the client's phone.

Implementation Assessment

Feature	Firmware Dev Effort	Hardware Changes	Primary Value
Gesture Profiles	Medium (I2C NDEF writing, tap tuning)	None	High user vers and personali
Data Logging	High (SRAM pass-through logic, FeRAM management)	None	Showcases engineering c provides uniq metrics.
Zero-Battery Wake	Low (Mostly relies on existing PMIC/Switch routing)	None	Fail-safe relial guarantees th effect.

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Which of these three features do you think aligns best with the project's core goals, so we can dive deeper into the firmware logic required to pull it off?

keep those ideas and continue extrapolating. consider whether this setup could function as a NFC key for my office, for example.

Can This Setup Function as an NFC Office Key?

The short answer is **yes, but with important security caveats.**

Whether this card can open your office door depends entirely on the type of access control system your office uses:

- **The "Simple" Yes (UID Reading):** Most basic or older NFC door locks, as well as many consumer smart locks, simply read the 7-byte Unique Identifier (UID) hardcoded into the NFC chip. The NT3H2211 has a standard NXP UID. You could easily ask your office manager to scan your card and enroll its UID into the system, treating it exactly like a standard plastic key fob.
- **The Security Reality (Cryptographic Handshakes):** Relying solely on a UID is fundamentally insecure, as UIDs are trivial to clone using a smartphone or a cheap RFID cloner. Modern enterprise systems (like HID iCLASS or MIFARE DESFire) require the chip to perform a cryptographic challenge-response at the hardware level. The NT3H2211 does not have this built-in cryptographic hardware.

- **The Custom Engineering Route:** Because your card has an MCU (AVR64EA28) connected to the NFC chip's SRAM pass-through, you could build your own secure lock. The door reader could send a random cryptographic puzzle (a nonce) to the card via NFC, your MCU could wake up, solve the puzzle using a stored secret key, and pass the answer back through the NFC chip to unlock the door.

Extrapolating New Features

Given the unique combination of solar harvesting, an accelerometer, FeRAM, and a dual-interface NFC chip, here are a few more advanced applications the build team could explore:

1. Hardware Authenticator (2FA / HOTP Token)

You can turn this business card into a physical two-factor authentication token.

- **How it works:** Instead of relying on time-based codes (which require a constantly running clock that might die if the card loses all power), you can use HMAC-based One-Time Passwords (HOTP). When you need to log into an account, you tap the card to your phone. The phone sends a counter via the NFC SRAM, the AVR MCU calculates the cryptographic hash, increments its internal counter (saving it to the FeRAM), and sends the login code back to the phone.

2. Secure Offline Vault

The 512kbit (64 Kilobyte) FeRAM chip offers a massive amount of storage for a business card—plenty of room for text data.

- **How it works:** You can store highly sensitive information, such as cryptocurrency recovery seeds, master passwords, or emergency medical info, in the FeRAM. This data remains completely isolated and offline. To access it, a companion app on your phone must send a specific password to the card via NFC. Only then will the MCU retrieve the data from the FeRAM and transmit it back to the phone.

3. "Light-Aware" Contextual Actions

Since the AEM10300 manages solar harvesting, the MCU can theoretically read the voltage or charging status to determine the ambient light level.

- **How it works:** The card can behave differently based on whether it is in a dark room (like a wallet or pocket) versus out in the bright sun. For example, if it detects it has just been pulled out of a dark pocket into a well-lit room, the MCU could trigger an automatic LED light sequence to impress the person you are about to hand it to, without requiring an accelerometer tap.

Given these possibilities—ranging from access control to cryptography and contextual awareness—do you foresee this card being used primarily as an interactive networking tool, or are you leaning toward using it as a personalized piece of utilitarian hardware?

consolidate these recommendations into a document i can upload to the build team

Here is the consolidated proposal document, structured for your build team's review. You can copy and paste this directly into your documentation platform or project management tool.

Feature Expansion Proposal: Solar Glow Business Card v4.0

Executive Summary

The hardware architecture of the Solar Glow Business Card v4.0 represents a highly capable embedded system. The current Bill of Materials—specifically the inclusion of the **NXP NT3H2211** dual-interface NFC chip, **AVR64EA28** microcontroller, **AEM10300** solar harvester, **ADXL367** accelerometer, and **512kbit FeRAM**—provides a foundation that extends far beyond a static digital business card.

ZIP +1

Because the NT3H2211 bridges the RF interface and the onboard I2C bus via a 64-byte SRAM pass-through, the card supports real-time, two-way communication between a scanning device and the MCU. This proposal outlines several firmware-driven features that require **zero hardware changes**, maximizing the versatility and utility of the existing v4.0 design.

Proposed Feature Enhancements

Category 1: Interactive & Contextual Networking

These features focus on enhancing the recipient's experience when receiving and scanning the card.

- **Gesture-Based Profile Switching**
  - **Components:** ADXL367 (Accelerometer), AVR64EA28 (MCU), NT3H2211 (NFC).
    ZIP
  - **Concept:** Utilizing the ultra-low-power tap/double-tap wake feature of the accelerometer, users can cycle through different digital profiles before handing the card to someone. A single tap configures the NFC NDEF record for a standard vCard; a double tap prompts the MCU to rewrite the NDEF record to point to a portfolio, demo, or GitHub repository.
  - **Feedback:** Onboard LEDs can flash to confirm the active mode.

- **Light-Aware Contextual Actions**
  - **Components:** AEM10300 (PMIC), AVR64EA28 (MCU), LEDs.
    ZIP
  - **Concept:** The MCU can read the voltage or charging status from the solar harvester to determine ambient light levels. If the card detects a transition from a dark environment (a pocket or wallet) to a well-lit room, the MCU can trigger an automatic LED sequence to impress the recipient without requiring a physical tap.

Category 2: Utility & Security

These features allow the owner to use the card as a personalized, secure multi-tool.

- **NFC Access Control / Office Key**
  - **Components:** NT3H2211 (NFC), AVR64EA28 (MCU).
    ZIP
  - **Concept:** For basic access control systems, the card's native 7-byte UID can be enrolled like a standard key fob. For advanced security, the MCU can facilitate a custom challenge-response protocol. A lock sends a cryptographic nonce via the SRAM pass-through; the MCU wakes, solves the puzzle using a stored secret, and passes the unlock token back.
- **Hardware Authenticator (HOTP Token)**
  - **Components:** NT3H2211 (NFC), AVR64EA28 (MCU), MB85RC512TY (FeRAM).
    ZIP
  - **Concept:** The card functions as a physical two-factor authentication key using HMAC-based One-Time Passwords (HOTP). Tapping the card to a companion app sends a counter via NFC. The MCU calculates the hash, increments the counter in the FeRAM, and returns the secure login code.
- **Secure Offline Vault**
  - **Components:** MB85RC512TY (FeRAM), NT3H2211 (NFC).
    ZIP
  - **Concept:** The 512kbit FeRAM offers 64 Kilobytes of storage—plenty of room for sensitive text data like master passwords or emergency information. This data remains completely offline. Access requires a companion app to send a password via NFC, prompting the MCU to retrieve and transmit the requested FeRAM data.

Category 3: Diagnostics & Reliability

These features leverage the system's memory and energy routing to ensure performance and track usage.

- **Environmental & Tap Data Logging**
  - **Components:** AEM10300, MB85RC512TY, NT3H2211.
    ZIP
  - **Concept:** The MCU periodically wakes to log solar harvesting levels and timestamps of accelerometer wakes or NFC field detections into the FeRAM. When the owner

scans the card, the phone queries the MCU, which streams an "Analytics Dashboard" back to the device.

- **Zero-Battery "Vampire" Wake**
  - **Components:** NT3H2211 (Energy Harvesting), TPS22917 (Load Switch), LEDs. ZIP
  - **Concept:** If the card's main energy buffer is completely depleted, the NT3H2211 can harvest up to 15mW of power directly from the scanning smartphone's RF field. This harvested energy can be routed through the TPS22917 load switch to briefly wake the MCU or directly pulse the LEDs, guaranteeing the card is never "dead" during a scan.

Implementation Assessment

Feature Focus	Required Firmware Dev Effort	Hardware Changes	Primary Value
Gesture Profiles	Medium (I2C NDEF writing, tap tuning)	None	High user vers and dynamic networking.
Light-Aware Actions	Low (ADC/PMIC polling logic)	None	Automated vi appeal.
Office Key / Access	Low to High (Depends on lock protocol)	None	Daily physical
Authenticator/Vault	High (Crypto libraries, FeRAM management)	None	Extreme secu personal data
Data Logging	High (SRAM pass-through, FeRAM wear leveling)	None	Showcases engineering c and metrics.
Zero-Battery Wake	Low (Relies on existing PMIC/Switch routing)	None	Fail-safe relial guarantees th

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